

— DESIGN — ARCHITECTURE — FABRICATION

AMD NSRI



Ideas, Designed into matter.

A working catalogue of facade systems, surface logic, and the fabrication intelligence that turns geometry into sheet, fold and finish.

01

Studio

02

Projects

03

Fabrication

— 01 / STUDIO

About The Studio

COMPUTATIONAL DESIGN · FAÇADE SYSTEMS · FABRICATION



FIG. 01 — SKETCH / SYSTEM / BUILT FAÇADE · AMD·NSRI 2026

A studio operates between computational design and fabrication — developing façade systems, digital tools and production-ready solutions from initial geometry to finished assembly.

DETAILS :

■ {01}

The Gap

Most facades fail between the render and the fabricator. The studio exists to close that gap — drawing for the cutting bed, not only the camera.

■ {02}

The Method

Perforation, structure and finish are reasoned together as one system. Grasshopper, Rhino and custom scripts keep that geometry buildable, screen to sheet.

■ {03}

The Build

Design is translated into buildable components — resolving geometry, material, detailing and production data for fabrication and assembly.

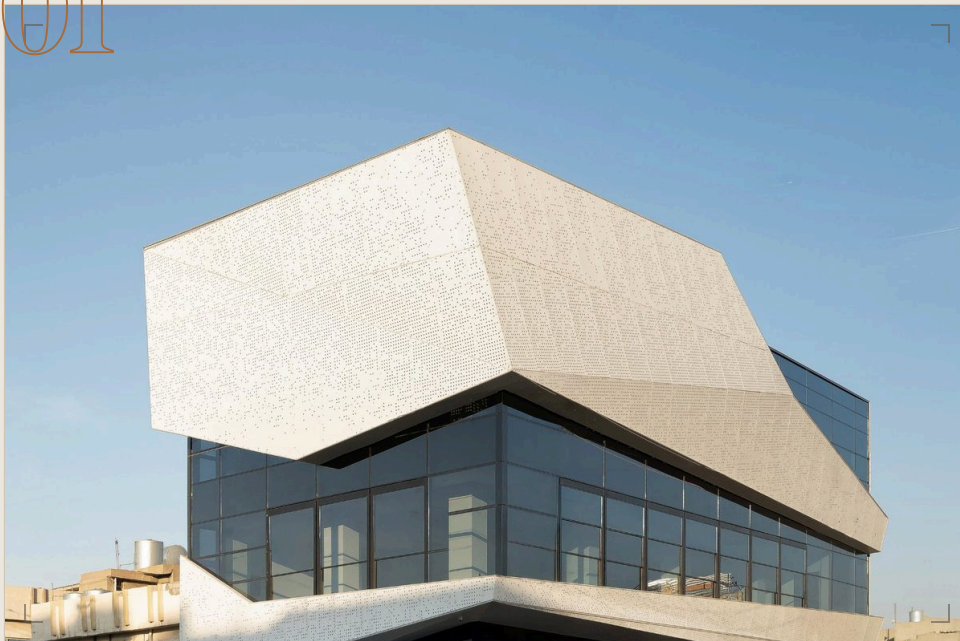
ARCHITECTURE / DESIGN / FABRICATION / TOOLS

02

◇ PROJECTS

Selected works.

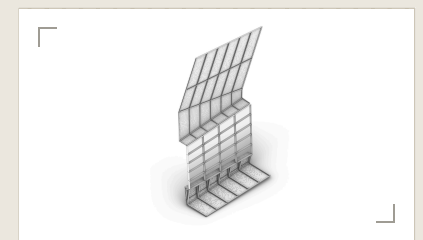
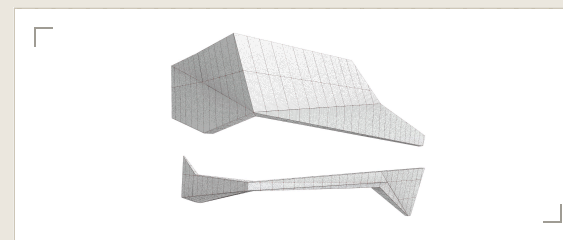
01



P·01 — Commercial

Restaurant Facade — Multiperforated Skin

◇ KIRKUK · IRAQ

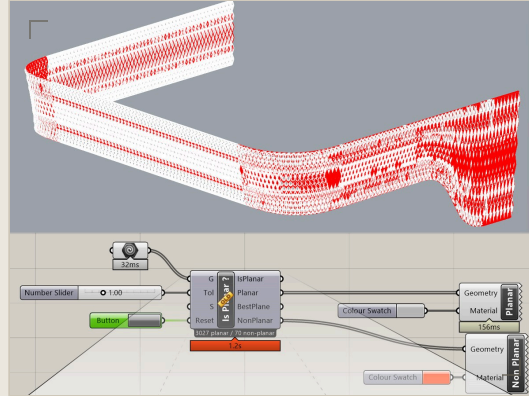


A gradient skin shaped by light and density.

Perforation density shifts across the façade, creating a controlled gradient of transparency, shade and privacy while giving the restaurant a distinct identity from the street.

FACADE · MULTIPERFORATED · DESIGN & FABRICATION

02



P.02 - Commercial · Computational Fabrication

Toyota SAS & Lexus Showroom

◇ SULAYMANIYAH · IRAQ

One façade, reduced to a controlled panel system.

The showroom façade was remodelled and rationalized computationally, reducing nearly 600 panels into a more efficient set of repeatable types. Grasshopper was used to group similar panels within defined tolerances, assign poses and automate unfolding for fabrication.

FACADE · PANEL RATIONALIZATION · UNFOLDING

03



P·03

National Hospital Facade

◇ BAGHDAD · IRAQ

A heart translated into a flowing perforated skin.

Variable circular perforations form a continuous wave across the façade, while heart-shaped motifs are embedded into the metal skin as a subtle reference to care and healthcare. Changes in hole size and density create movement, transparency and a distinct identity for the hospital.

HEALTHCARE · HEART MOTIF · GRADIENT PERFORATION ·
PARAMETRIC FACADE

04



P·04 — Healthcare

Baghdad Private Hospital

◇ BAGHDAD · IRAQ

A fluid skin that filters light and calms the street.

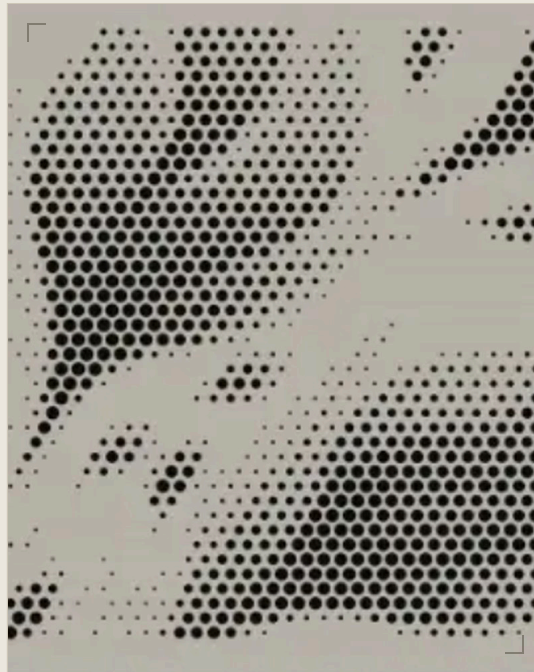
Baghdad Private Hospital explores a contemporary healthcare identity through a fluid parametric façade. Perforated aluminum panels filter natural light, provide privacy, and create a dynamic architectural expression that changes throughout the day, combining performance with a calm, welcoming presence.

HEALTHCARE · PARAMETRIC FACADE · PERFORATED ALUMINUM

05



01 IMAGE



→ 02 PERFORATED PANEL

P·05 — Digital tool

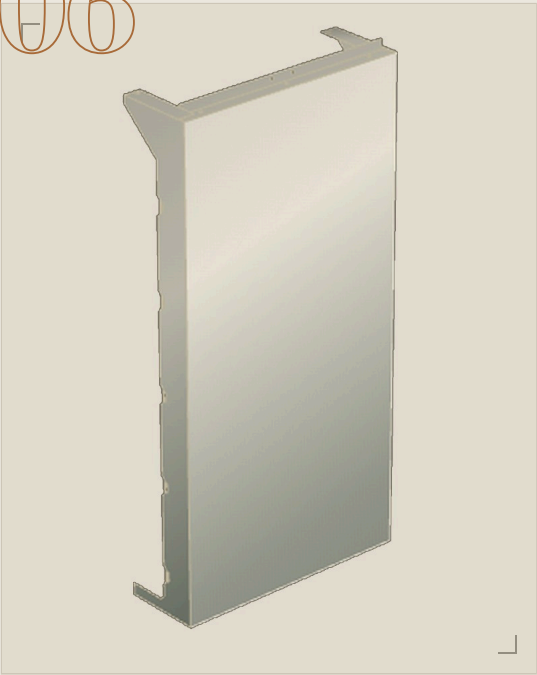
IPunch — Image to Perforation

Images become fabrication-ready perforation patterns.

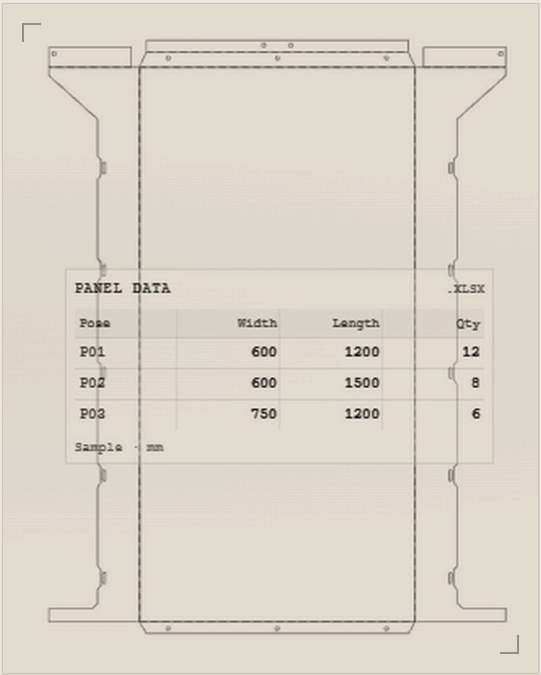
IPunch converts grayscale imagery into variable-diameter perforation fields, translating pixel brightness into controlled hole sizes and density. The tool allows adjustment of diameter range, spacing and grid orientation while generating a live preview and fabrication-ready output.

IMAGE SAMPLING · VARIABLE PERFORATION · DXF OUTPUT

06



01 3D PANEL



→ 02 FLAT PATTERN

P-06 - Fabrication service

Sheet Metal Unfolding

Complex panels, flattened for the cutting bed.

An unfolding service for complex sheet metal panels. Submit your files — DWG, PDF, image, or any usable reference — and receive accurate flat cutting patterns with correct k-factor and bend allowance applied. Output is DWG-ready for laser cutting and press brake, plus STEP for CNC verification.

FLAT PATTERNS · K-FACTOR · DWG · STEP

◇ FABRICATION

03 Six ways to work a metal surface.

A facade can be opened, stretched, cut, pressed, folded or drawn from scratch. Each system is a different way to **let light and air through metal** — and a different fabrication logic behind it.

PLATE 02 - SYSTEMS

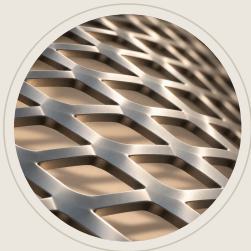
SIX FAMILIES



Perforated Metal

PUNCHED HOLE FIELD

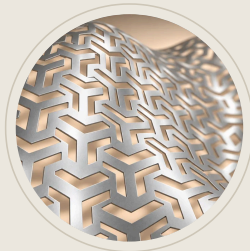
OA 5 - 62%



Expanded Mesh Metal

SLIT & STRETCHED LATTICE

OA ≈ 50%



Pattern Cut Metal

CNC GEOMETRIC VOIDS

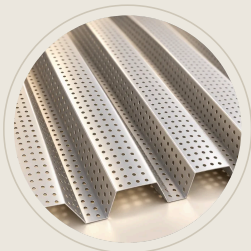
CUSTOM VOIDS



3D Surfaces

PRESSED RELIEF & DEPTH

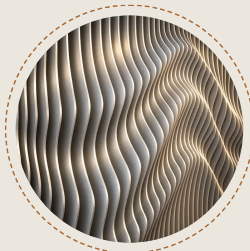
DUAL-DEPTH



Free Section

FOLDED BESPOKE PROFILE

FOLDED



Custom Design

ONE-OFF, DRAWN TO BRIEF

TO BRIEF

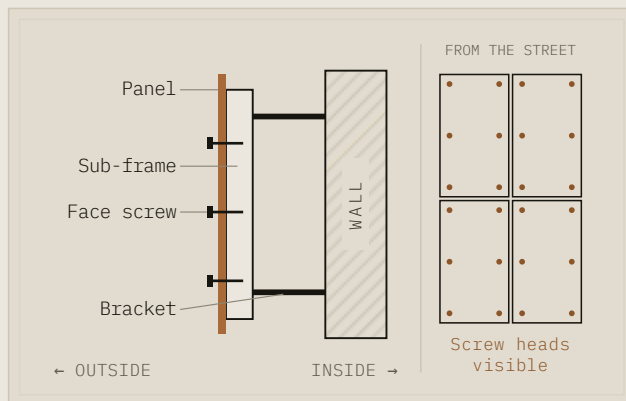
— DETAIL · ENVELOPE

How the panel meets the structure.

The carrier is the same idea everywhere — **bracket, profile, panel**. What changes is the gap behind, the joint between, and how the panel is held. Three conceptual approaches:

PLATE 03 — CONNECTIONS

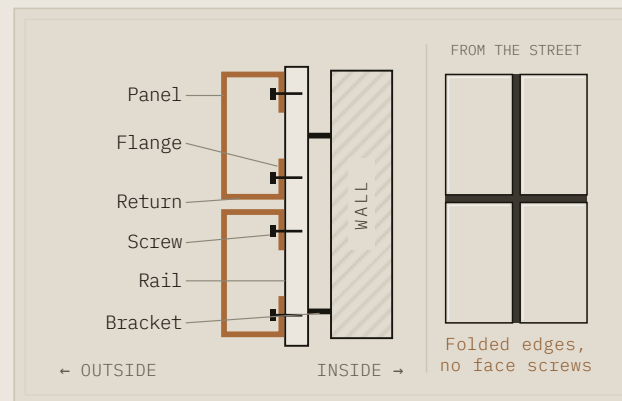
THREE APPROACHES



D.01 Direct Fixed

Panel sits tight to a box-profile sub-frame and is fixed straight through. Fewest parts, lowest cost — the default where a flat, rigid face is enough.

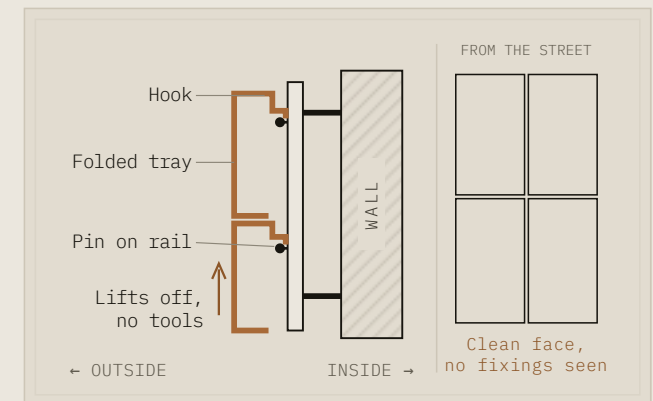
FACE / REAR FIX · MINIMAL CAVITY · ECONOMICAL



D.02 Monoblock Fixed

Each panel edge is folded twice — a return toward the wall, then a flange — and screwed to the sub-frame rail through that last fold. Fold depths are set per project; the folds stiffen the panel and keep every fixing off the face.

DOUBLE-FOLDED EDGE · FLANGE-SCREWED · RIGID PANEL



D.03 Cassette Hook-on

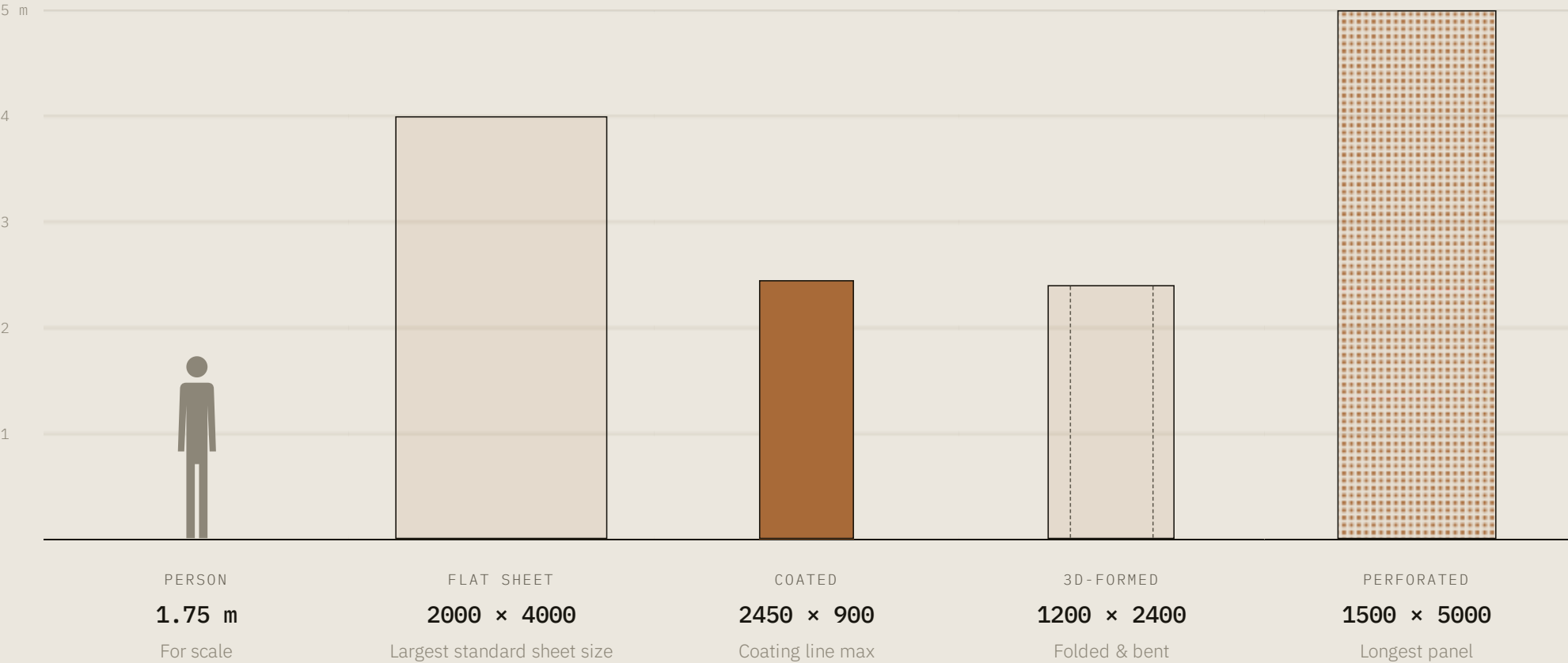
Panel folded into a tray; the return hooks over a carrier rail. No visible fixings, and any panel lifts off on its own — the choice when the face must be clean and serviceable.

CONCEALED FIXING · FOLDED TRAY · DEMOUNTABLE

Size Limits

— LARGEST PANEL PER PROCESS · TO SCALE · MM

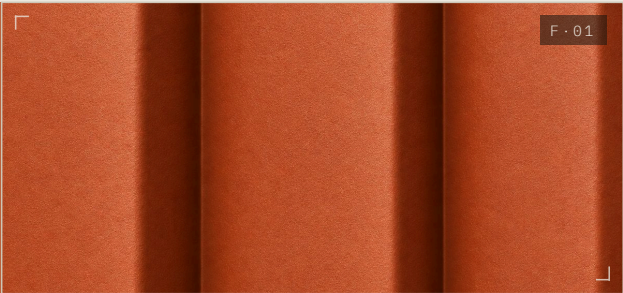
How big can a panel be? Each shape is the largest piece one process can take — drawn to scale, longest side up, next to a person.



— FINISH · MATERIAL

Surfaces you can specify.

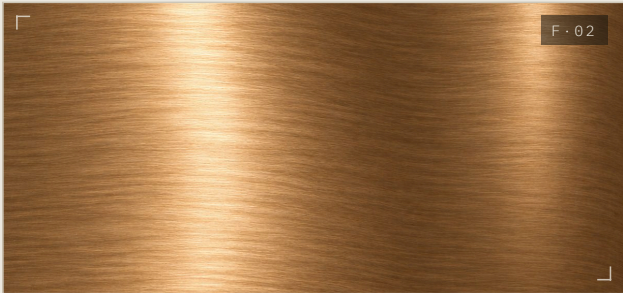
Some surfaces are applied to the metal, some are the metal itself — **a coating, a printed texture, or a material left to age.** Together they set how a facade reads, and how long it holds.



Powder Coating

SOLID COLOUR

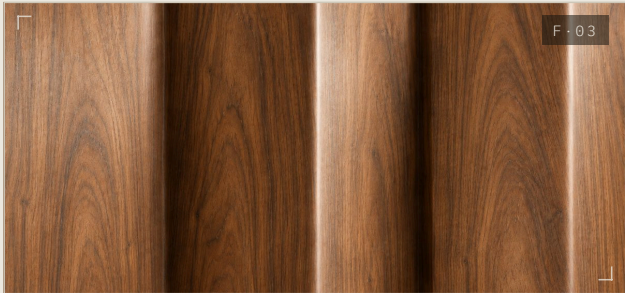
- Matte
- Gloss
- Texture



Anodizing

METALLIC TINT

- Matte
- Brushed
- Shiny
- Satin



Wood Effect Transfer

TIMBER GRAIN

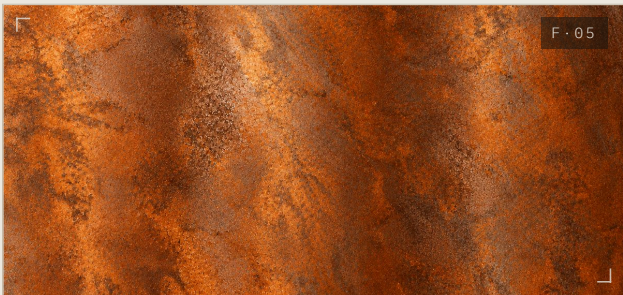
- Oak
- Walnut
- Teak



Stainless Steel

POLISHED REFLECTION

- 304
- 316
- 430



Corten

WEATHERED PATINA

- Pre-aged
- 6 months
- 1 year
- 2-3 years



Galvanize

PROTECTIVE ZINC

- Hot-dip
- Pre-galvanized

— FINISH · COATING

Three classes, one long decision.

QUALICOAT sets three durability classes for powder coating, each with an **AAMA 2600-series equivalent**. Same single-coat application — the difference is the chemistry, and what it buys in years.

01

CLASS · 1
QUALICOAT

Polyester

10_{YR}

60MM · WIDE COLOUR RANGE
STANDARD RESIN · FLORIDA 1YR

AAMA 2603

02

CLASS · 2
QUALICOAT

Super Durable PE

20_{YR}

60MM · WIDE COLOUR RANGE
MODIFIED POLYESTER · FLORIDA 5YR

AAMA 2604

03

CLASS · 3
QUALICOAT

Durasol PVF3

30_{YR}

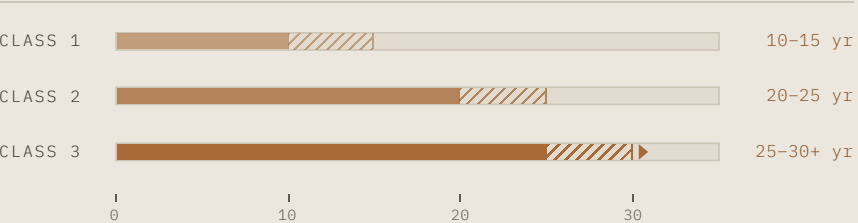
50MM · LIMITED COLOUR RANGE
FLUOROPOLYMER (PVDF) · FLORIDA 10YR

AAMA 2605

WEATHERING PERFORMANCE

	C1	C2	C3
UV Resistance	◆◆◆◆	◆◆◆◆◆	◆◆◆◆◆◆
Colour Retention	◆◆◆◆◆	◆◆◆◆◆	◆◆◆◆◆◆
Chalk Resistance	◆◆◆◆◆	◆◆◆◆◆	◆◆◆◆◆◆
Gloss Retention	◆◆◆◆◆	◆◆◆◆◆	◆◆◆◆◆◆

ARCHITECTURAL LIFESPAN



SOLID = MINIMUM · HATCHED = EXPECTED RANGE
MANUFACTURER GUARANTEE 10 · 20 · 30 YR – SAME ORDER

SALT TEST · ASTM B117

1000–2000 hr

Class 1

COASTAL DURABILITY

Medium → Very high

C1 → C3

ADHESION · ISO 2409

GT0 across all classes

No class trade-off here

— COLOPHON

AMD NSRI.



CONCEPT



BUILT

STUDIO

Ahmed Alnaseri

Architect · Computational Design

Istanbul · Türkiye

ahmed@amdnsri.com

[+90 537 872 64 52](tel:+905378726452)

CATALOGUE

Edition 01 – 2026

Design & Fabrication Index

amdnsri.com

SCAN



SCAN TO VISIT
AMDNSRI.COM